

ABOUT CELLS



In this worksheet you will start to learn about cells. There are many new words to learn about cells.

It was in 1665 that Robert Hooke first used a very simple microscope to look at bark. He discovered that the bark was made up of little boxes. These boxes looked like the 'cells' that priests lived in. His discovery started a whole new idea or 'theory' about the building blocks of living things.

Learning about cells means there are many new words to learn.

Learning the meanings and spellings of the new 'cell' words will make the work about cells easier to understand.

Look at the table of words and meanings. Then follow the instructions to help learn the words.

special words

microscope
cells
magnification
cell theory
cytoplasm
nucleus
mitochondrion
eyepiece lens
microscope tube
coarse focus knob
fine focus knob
objective lens
stage
mirror
base

Word	Meaning
Microscope	A tool that helps us to see small things such as cells that we cannot see with our eyes.
Magnification	How many times bigger the object looks than it really is. Strong microscopes magnify objects many times.
Cell theory	Some ideas or rules about cells.
Cytoplasm	A living, jelly-like material that makes up the inside of cells. It has lots of chemicals and cell parts in it.
Nucleus	The control centre of the cell. It is a bit like the brain of the cell.
Mitochondrion	Small parts of the cell that make the energy for the cell to work.

To do



- Write out each word twice in the spaces under the word. Make sure that you copy the word out carefully so that you learn to spell it properly.
- Ask your teacher how to say each word.

nucleus

microscope

cell theory

cytoplasm

mitochondrion

magnification

The cell theory tells us that:

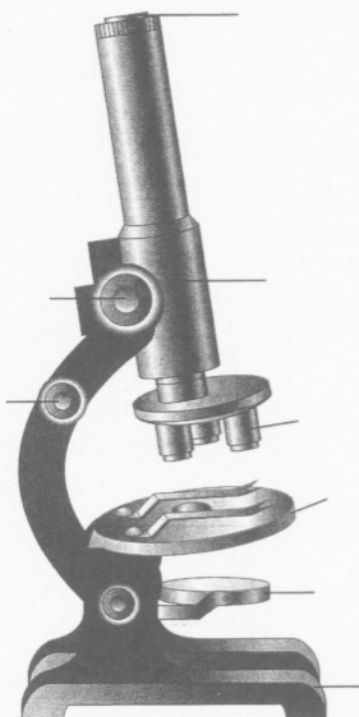
- All living things are made up of cells or come from things made by cells.
- Cells are the building blocks and working units of living things.
- We look at cells through a microscope.

To do



You need to know about some of the parts of the microscope so that you can use one properly.

- 1 Use a textbook and the word list to label the parts of the microscope in the spaces given. Your teacher will explain what each of these parts does.



Word list:

eyepiece lens
microscope tube
coarse focus knob
fine focus knob
objective lens
stage
mirror
base

- 2 Fill in the gaps using the words: microscope, through, holds, seen, bigger, lens, object.

Name of part	What it does
Eyepiece lens	Bends the light to make the object look _____
Microscope tube	Light travels _____ it to the eyepiece
Coarse focus and fine focus knobs	Moves the lens up or down so that the object can be _____ clearly
Objective lens	Can be changed so that the o _____ can be magnified more or less
Stage	H _____ the slide with the object you are looking at in place
Mirror	Makes light bounce through the slide into the l _____
Base	Holds the m _____ up

 In this worksheet you will learn more about cells, especially animal cells.

There are many different types of cells. Their shape depends on the job they have to do and so they often look a bit different to one another. To make cells easier to understand you will need to learn some more words. Some of these words are difficult to spell and to say so you can practise them now.

special words

cells
cell membrane
nuclear membrane
vacuole
ribosome
chromosome
cytoplasm
nucleus
mitochondrion
mitochondria

Word	Meaning
Cell membrane	A thin boundary around the cell that keeps things inside cells and lets some needed things into the cells.
Nuclear membrane	A very thin boundary around the nucleus of the cell.
Vacuole	A storage area of the cell that is full of fluid.
Ribosomes	Little round parts of cells that make chemicals called proteins. Proteins are needed so that we can survive and grow.
Chromosomes	Chemicals in the nucleus of the cell that carry all the information that we get from our parents.

To do



- Write out each of the words twice in the spaces that you have been given.

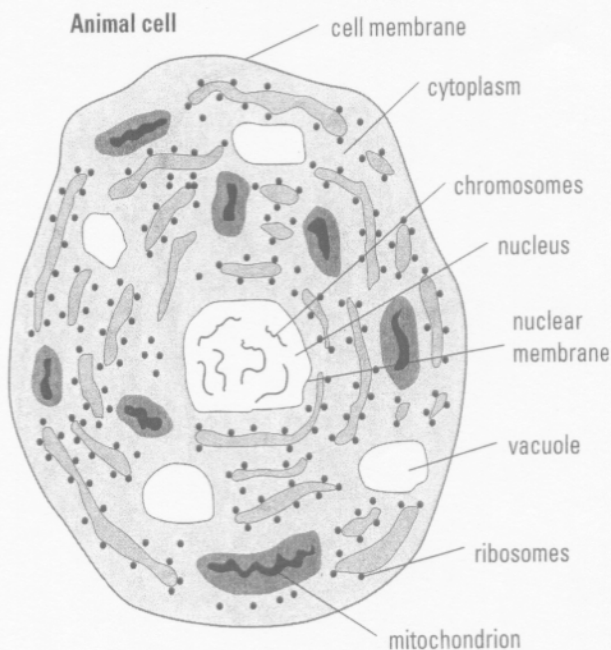
cell membrane

nuclear membrane

vacuole

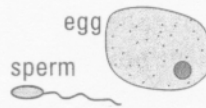
ribosomes

chromosomes



Sex cell

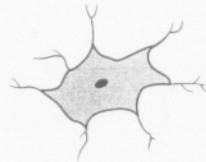
Male sex cells are called sperm. Female sex cells are called eggs or ova. Sex cells combine during reproduction to form a new organism.

**Muscle cell**

A muscle cell can contract (shorten) to produce movement.

**Brain cell**

A brain cell makes electrical messages which it passes to other cells through its many connections.

**Motor nerve cell**

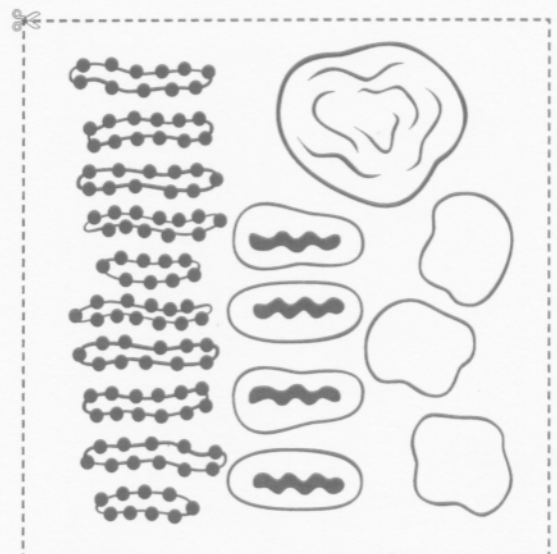
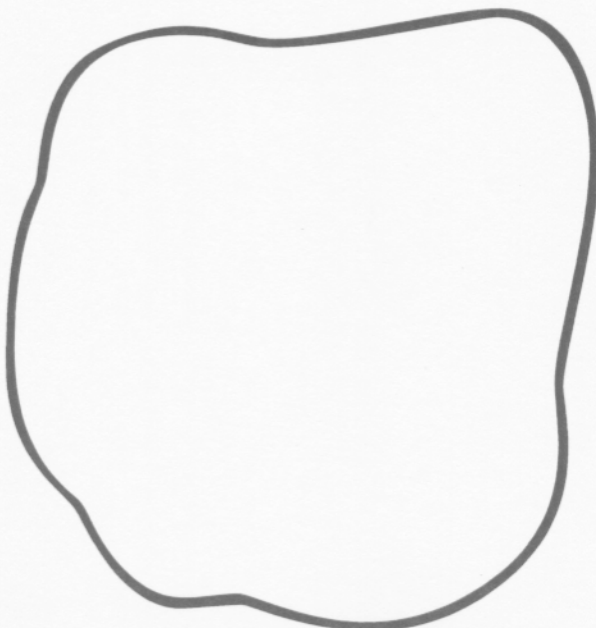
A motor nerve cell controls movement. It passes a message from the brain to a muscle.

**White blood cell**

White blood cells help to fight disease. Some move about in the body to 'swallow up' bacteria.

**To do**

- 2 Using the labels on the animal cell as a guide, label the parts of all the cells on this page.
- 3 Now make your own animal cell.
 - Cut out the cell parts from below.
 - Colour in the parts so that they will be easy to see. Make sure that each of the cell parts has its own colour.
 - Paste the parts onto the page using the labelled animal cell as a guide.





In this worksheet you will learn about plant cells.

Plant cells have the parts that you have learnt about for animal cells and two extra parts that are *not* found in animal cells. These parts are the cell wall and chloroplasts.

To make cells easier to understand, once again you will need to learn some new words. Some of these words are difficult to spell and to say so you can practise them.

Word	Meaning
Cell wall	A boundary around the cell that gives the plant cells a definite shape. Plant cell walls help to give plants shape because plants do not have bones or skeletons like many animals.
Chloroplast	Green parts of plant cells that carry out the chemical reaction called photosynthesis. Photosynthesis turns light energy from the sun into chemical energy (carbohydrates) that animals can use for energy to survive when the plants are eaten.

special words

plant cell
animal cell
cell wall
chloroplast
cell membrane
nuclear membrane
vacuole
ribosome
chromosome
cytoplasm
nucleus
mitochondrion
mitochondria

To do



- Write out each of the words twice in the spaces that you have been given.

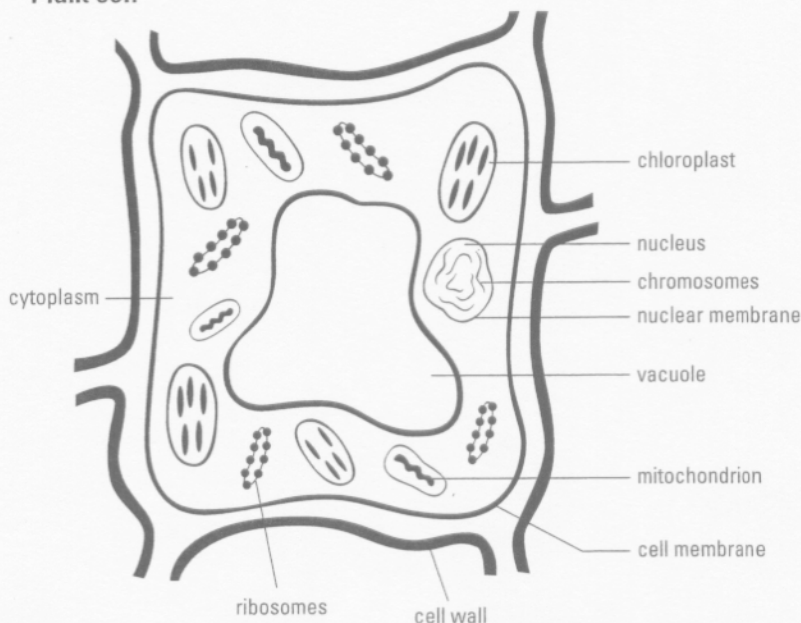
cell wall

chloroplast

Cell membrane, nuclear membrane, vacuole, ribosomes, chromosomes, cytoplasm, nucleus, mitochondrion (many mitochondria), cell wall and chloroplast are all parts of plant cells.

On this page and the next are some drawings that show what plant cells can look like and what the parts of plant cells look like. Cells from different plants look different because each cell has a different job.

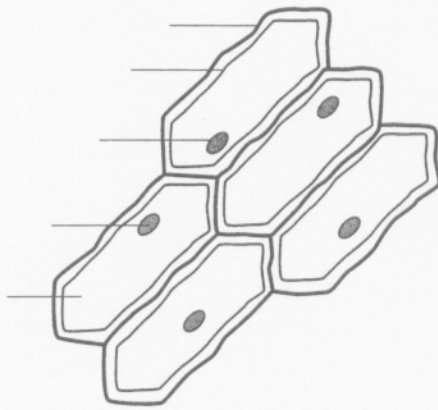
Plant cell



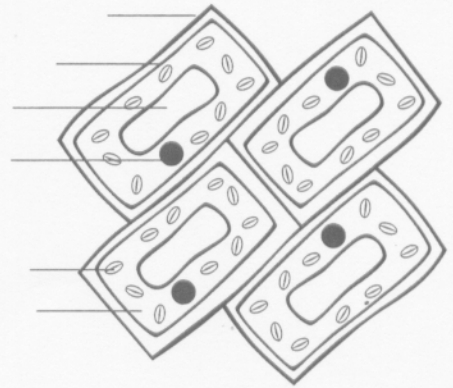
To do



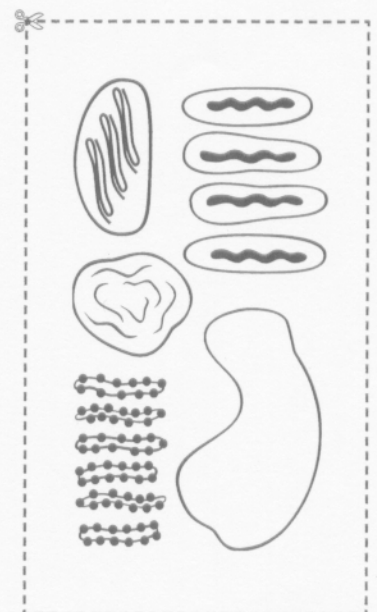
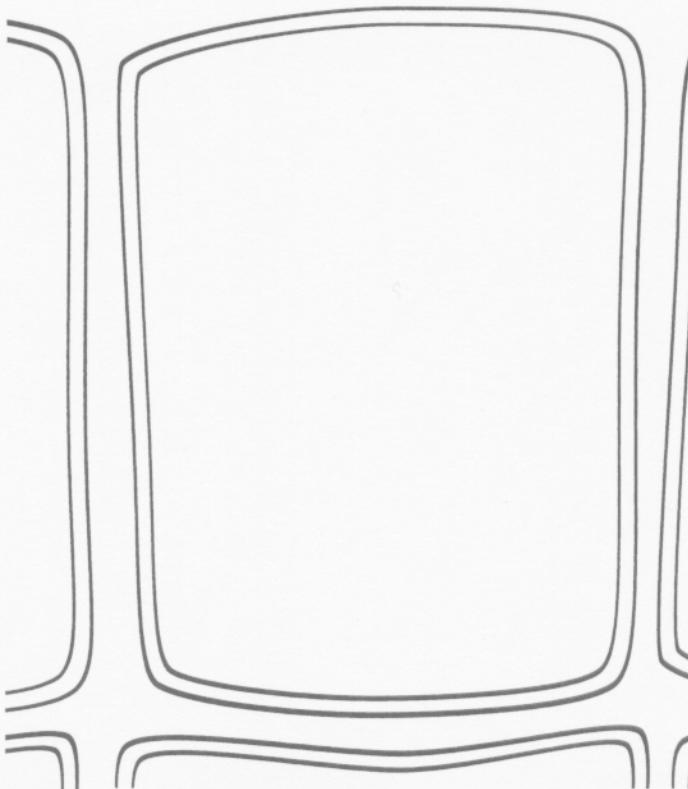
- 2 Using the labels on the plant cell as a guide, label the parts of all the cells on this page.
- 3 Now make your own plant cell.
 - Cut out the cell parts from below.
 - Colour in the parts so that they will be easy to see. Make sure that each of the cell parts has its own colour.
 - Paste the parts onto the page using the plant cell on this page as a guide.



Onion cells



Pondweed cells





In this worksheet you will learn how different cells working together make up an organism.

In any plant or animal there may be millions of different cells. Each cell will carry out its own job. All the cells working together will make an organism.

In worksheets 13.2 and 13.3 you learnt about some different animal and plant cells.

The animal muscle cell can shorten so that bones can move.

- One muscle cell by itself is not strong enough to move a bone, but many muscles cells together would be.
- Many muscle cells together are called muscle tissue.

Tissue is a group of cells working together.

- The cells all look the same and do the same job.
- A group of nerve cells together would be called nerve tissue.

A group of tissues working together makes an organ.

- The human lungs are an example of an organ.
- The lungs are made up of: muscle tissue, blood vessels, and nerves all working together to help us breathe in and out.

Many organs working together are called a system.

- An example of this is the circulatory system.
- The circulatory system is made up of the heart, blood vessels, and the blood.

All the systems working together make up an organism!



special words

organism
muscle cell
muscle tissue
tissue
nerve tissue
organ
system
circulatory system

To do



1 Finish the sentences below using the word list: system, living things, organ, tissue.

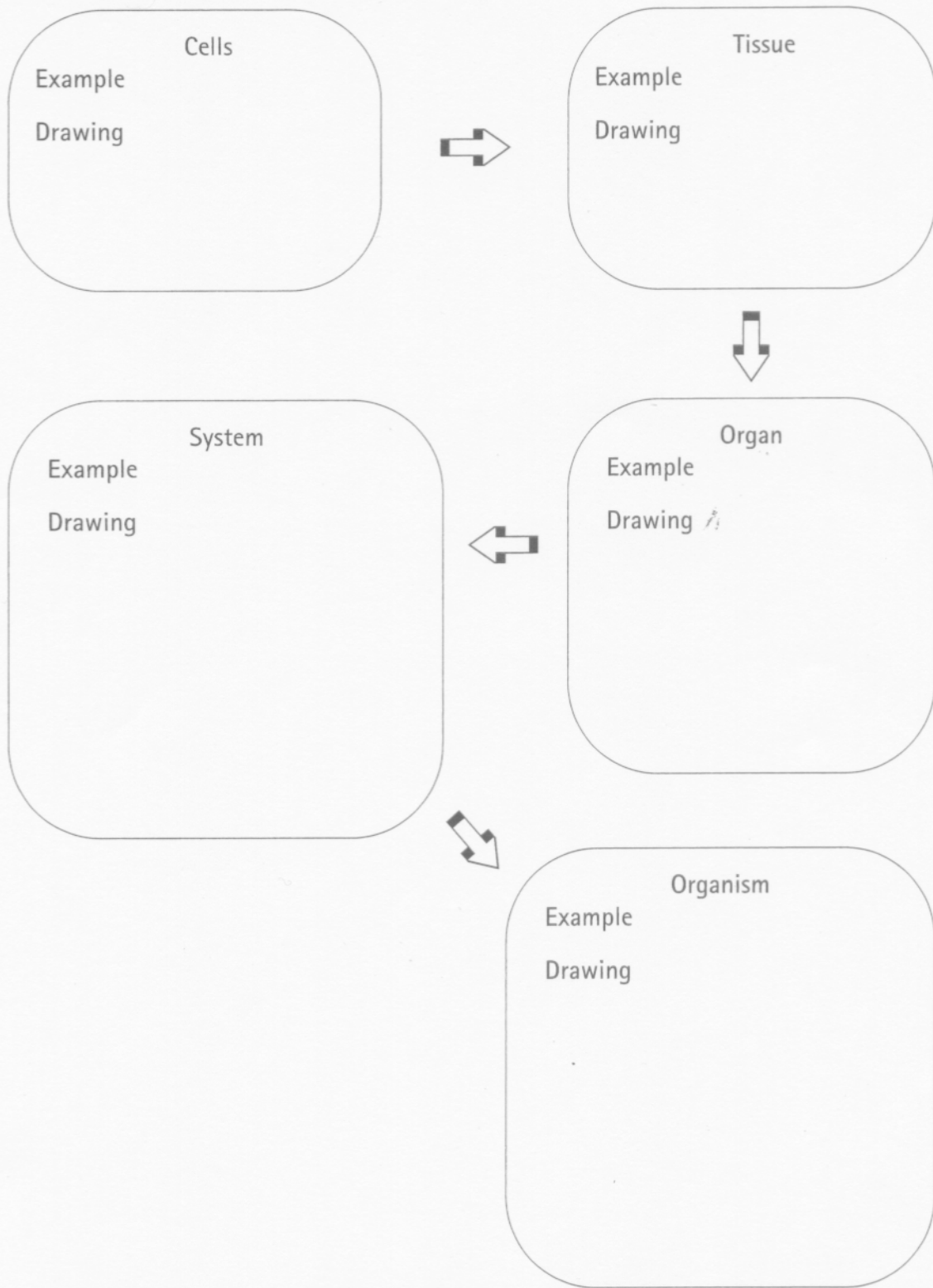
- A cell is the building block for all _____.
- A group of cells that all work together to do a special job are called _____.
- Many tissues working together make up an _____.
An example of this is the lungs.
- A lot of different organs working together make up a _____.
The circulatory system is an example of this.

To do



- 2 Complete the flowchart below so that it shows how cells fit together to make an organism.
- 3 For each section give an example and where you can draw a picture.

Flowchart of cells to organism





Here is a word puzzle to help you remember all the new words that you have learnt about cells.

To do



Use the list of clues and lists of words on worksheets 13.1, 13.2, 13.3 and 13.4 to help you find the words in the puzzle.

P	O	P	Y	G	J	H	F	W	O	X	Z	F	C	V	U	Z	E
M	M	A	G	N	I	F	I	C	A	T	I	O	N	C	M	M	H
K	I	X	O	I	E	L	R	E	V	Y	J	R	Y	H	Q	P	C
F	T	X	I	W	T	P	S	L	U	W	O	A	P	L	N	R	E
X	O	R	G	A	N	A	Q	L	K	B	L	S	O	O	U	B	L
M	C	R	Y	Y	R	Q	T	T	E	I	T	F	L	R	C	W	L
N	H	I	G	J	Q	U	D	H	I	Y	P	J	S	O	L	V	M
G	O	C	V	A	Q	L	C	E	L	L	V	U	L	P	E	C	E
T	N	M	R	J	N	B	D	O	C	Y	T	O	P	L	A	S	M
Z	D	I	C	C	J	I	J	R	Y	S	M	Q	R	A	R	Y	B
G	R	C	M	Y	C	H	S	Y	D	Y	Z	F	H	S	M	I	R
D	I	R	I	B	O	S	O	M	E	S	S	B	J	T	E	D	A
T	O	O	F	H	H	Q	X	O	Z	T	L	W	O	L	M	Z	N
X	N	S	W	N	U	H	K	D	S	E	I	S	X	L	B	J	E
H	W	C	H	R	O	M	O	S	O	M	E	S	U	D	R	Y	H
M	G	O	Z	E	E	H	N	U	C	L	E	U	S	U	A	Z	C
S	E	P	I	Z	S	L	A	W	N	B	J	I	Q	U	N	N	T
Y	C	E	L	L	W	A	L	L	V	A	C	U	O	L	E	O	L

List of clues

- Some ideas or rules about cells.
- A very thin boundary around the nucleus of the cell.
- A storage area of the cell that is full of fluid.
- The basic building blocks of all living things.
- Many cells working together to do the same job.
- Many tissues working together to do the same job.
- Many organs working together to do the same job.
- All the systems together.
- A tool that helps us to see small things, such as cells, that we cannot see with our eyes.
- How many times bigger the object looks than it really is. Strong microscopes magnify objects many times.
- A living, jelly-like material that makes up the inside of cells. It has lots of chemicals and cell parts in it.
- The control centre of the cell. It is a bit like the brain of the cell.
- Small parts of the cell that make the energy for the cell to work.
- A thin boundary around the cell that keeps things in cells and lets some things that are needed into the cells.
- Little round parts of cells that make chemicals called proteins. Proteins are needed so that we can survive and grow.
- Chemicals in the nucleus of the cell that carry all the information we get from our parents.
- A boundary around the cell that gives the plant cells a definite shape. Plant cell walls help to give plants shape because plants do not have bones or skeletons like many animals.
- Green parts of plant cells that carry out the chemical reaction called photosynthesis. Photosynthesis turns light energy from the sun into chemical energy (carbohydrates) that animals can use for energy to survive when plants are eaten.